

Medicine Search Rules, Examples, and Current Evaluation

What the system handles, where it stops, and how we tested it

The short version

The system combines ordinary typo search, bounded OCR/handwriting rules, explicit visual-gap search, and optional product details. It does not silently merge every similar-looking letter: exact names, ambiguous families, and conflicting product details are protected and always require confirmation.

Current source authority	A5 c5faefb2...; A6 cedf1fce...; product reranker 03780539...; API 752399ee...
Catalog	25,066 product records; 17,476 exact medicine families
Evaluation rule	Accuracy, safety, and developer regression are reported separately
Deployment claim	Local candidate evidence only; this document does not claim clinical validation or public deployment

The accompanying evidence folder contains the exact test rows, stable family/product labels, generation manifests, selected per-case outcomes, compact summaries, and preserved failure history.

Contents

1 What the system can handle	3
2 How one query becomes a result	3
2.1 Normalization keeps different evidence separate	3
2.2 Retrieval and consensus rules	3
2.3 API and interface boundaries	4
3 Name reading rules	4
3.1 The direct OCR and handwriting registry	4
3.2 All weighted spelling groups—not keyboard rules	5
3.3 All OCR digits, ligatures, and phonetic rewrites	5
3.4 Bounded long phonetic collapse plus one ordinary error	5
3.5 Skeleton, phonetic, and chain keys	6
3.6 The 27 explicit source aliases	6
3.7 Budgets that keep the OCR rules bounded	7
3.8 How a family enters the candidate pool	7
3.9 Edit costs, prefilter, and full family score	8
3.10 When a candidate may move up: every promotion family	8
3.11 The 17 exact-chain promotion policies	10
4 Visual-gap rules	11
4.1 Four matching stages	11
4.2 What the new visual benchmark actually tests	12
5 Product-detail rules	12
5.1 Numeric evidence	13
5.2 Form, route, release, and container	13
5.3 Family admission and selection	13
5.4 Numeric aliases and very short prefixes	14
6 Safety rules around every result	14
7 The new test sets	14
7.1 Hard name-reading standard: five frozen files	14
7.2 What “hard” means in this suite	15
7.3 Worked test cases: what one row contains and how it is judged	15
7.4 Generation protocol without result leakage	16
7.5 Other evidence streams, kept separate	17
7.6 The row recipes, in plain language	17
8 Current-source results	17
8.1 New hard name-reading benchmark	18
8.2 Accuracy benchmarks	18
8.3 Safety and strict selection	18
8.4 Developer regression gates—reported separately	19

8.5 Fair comparisons only 19

9 What failed, what changed, and what was rerun 19

9.1 Hard name benchmark: protocol failure, then runtime failure 19

9.2 Strict product benchmark: 435/440 first run 20

9.3 Visual benchmark: 422/424 first run 20

9.4 Failure policy for every future rule 20

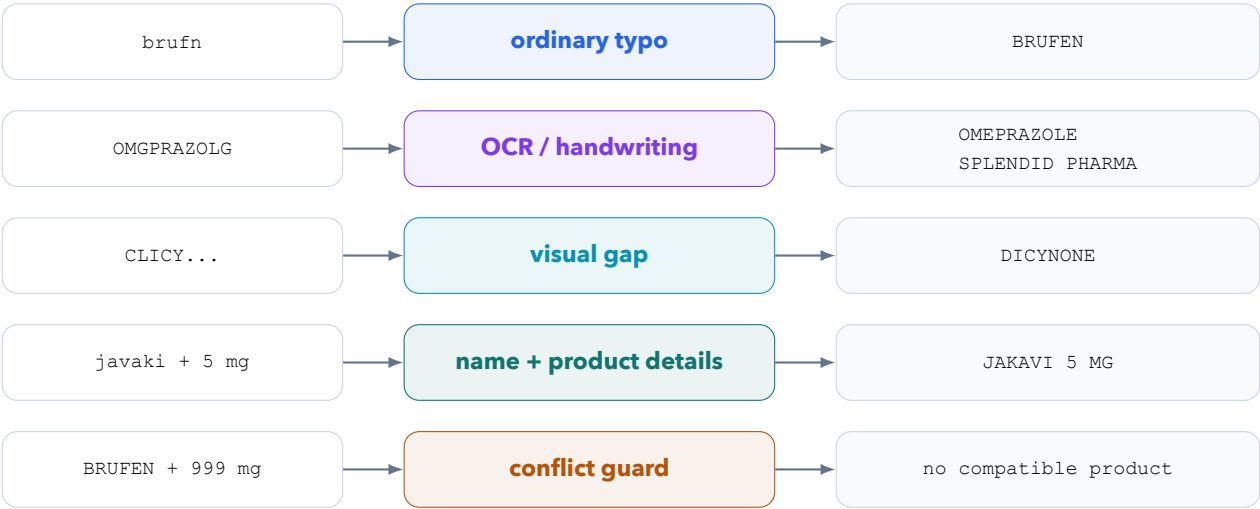
10 Open team decisions 20

A Rule-to-test checklist for contributors 20

B Current authority identifiers 21

C Dataset and run identifiers 21

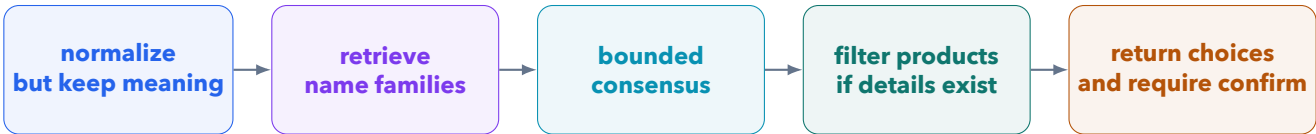
1 What the system can handle



How to read the rest of this handbook

Every rule is shown as **observed text** → **allowed catalog interpretation**, followed by a positive example and the safety condition that prevents it from becoming unrestricted fuzzy search.

2 How one query becomes a result



2.1 Normalization keeps different evidence separate

Input layer	Active normalization	Example and boundary
Medicine name	Case, repeated spaces, punctuation, Arabic digits, common Arabic letter variants, elongation and diacritics are normalized; a compact key removes separators.	BRU-FEN and brufen share a lookup surface, but an explicit visual marker is detected before punctuation cleanup.
Product details	Arabic digits and decimal/thousands separators, micro symbols, dotted units, release abbreviations, multipacks and known aliases are normalized before parsing.	An Arabic-digit one-point-five MG dose normalizes to 1.5 MG; 20 µg ↦ 20 UG; malformed or zero doses fail closed.
Identity	Names use exact compact base-family keys; selected products use stable catalog IDs. Broad variant groups are display metadata only.	Two products with the same printed name remain separate if their catalog IDs, manufacturers, or prices differ.

2.2 Retrieval and consensus rules

Algorithm 5 builds the main candidate order from exact and alias evidence, prefix/suffix and family-head indexes, tokens and character n-grams, bounded deletes and edit distance, phonetic keys, skeleton keys, and the OCR/ligature rules in the next section. Explicit visual-gap syntax takes its own branch.

Algorithm 6 then compares ten rankings: Algorithm 5 plus Jaro–Winkler, Match Rating, WRatio, NYSIIS, BM25+character trigrams, SymSpell edit-distance three, token-sort similarity, Soundex, and character-bigram Dice.

Consensus rule	Current behavior
Base order	Algorithm 5 remains the base ranking. The Algorithm 6 rank-1 promotion gate is disabled.
External top-20 fill	If Algorithm 5 already supplies 20 rows, at most one absent family may replace the tail. If it supplies fewer rows, every vacant slot may be filled. Each external family still needs at least three retrievers, Levenshtein similarity $\geq .55$, and Jaro–Winkler $\geq .85$.
Learned OCR costs	Recorded as a diagnostic similarity only; they do not independently reorder results.
Decision state	Every row is low-confidence decision support with <code>confirmation_required=true</code> .

2.3 API and interface boundaries

The API accepts 1–120 characters for the medicine query, up to 120 for optional product details, and a result limit from 1 to 20. It hydrates display fields by stable product ID first. The interface cache keeps raw visual-marker semantics, so `PANA DOL` and `PANA...DOL` are different requests. The static browser fallback is a separate search implementation and is never reported as Python Algorithm 6 parity.

3 Name reading rules

3.1 The direct OCR and handwriting registry

These are the explicit observed-text-to-catalog mappings. Direction matters: `D` seen on paper is not always given the same cost as catalog `D` being read as two characters.

Observed $\mapsto$ cata-log	Cost	Positive example	Main guard
I $\mapsto$ E	.45	OMIPRAZOLI $\rightarrow$ two I/E reads $\rightarrow$ OMEPRAZOLE SPLENDID PHARMA	Literal exact names stay protected
E $\mapsto$ I	.45	PRESOLINEBLUE $\rightarrow$ one E/I read $\rightarrow$ PRISOLINE BLUE	Original span consumed once
Y $\mapsto$ E	.45	ROVADENY $\rightarrow$ one Y/E read $\rightarrow$ ROVADENE	No transitive group merge
E $\mapsto$ Y	.45	PERAZIDE $\rightarrow$ one E/Y read $\rightarrow$ PYRAZIDE	Globally unique exact family
I $\mapsto$ Y	.45	OXILOZIME $\rightarrow$ one I/Y read $\rightarrow$ OXILOZYME	Bounded operation count
Y $\mapsto$ I	.45	SALMOCALCYN $\rightarrow$ one Y/I read $\rightarrow$ SALMOCALCIN	Bounded operation count
E $\mapsto$ G	.60	INNOVAEAR $\rightarrow$ one E/G read $\rightarrow$ INNOVAGAR	Direct retrieval only
G $\mapsto$ E	.60	OMGPRAZOLG $\rightarrow$ two G/E reads $\rightarrow$ OMEPRAZOLE SPLENDID PHARMA	Tail retrieval, not a global edit discount
CL $\mapsto$ D	.40	CLICYNONE $\rightarrow$ CL/D joined stroke $\rightarrow$ DICYNONE	Variable-length direct rewrite
D $\mapsto$ CL	.55	BACTIDOR $\rightarrow$ D/CL split stroke $\rightarrow$ BACTICLOR	Higher reverse cost
AL $\mapsto$ D	.45	ALEPIDERM $\rightarrow$ AL/D joined stroke $\rightarrow$ DEPIDERM	Global unique best family
D $\mapsto$ AL	.70	DKDINE $\rightarrow$ two D/AL reads $\rightarrow$ ALKALINE	Narrow depth-two exception

D / EL is not an implemented rule

The team specifically raised `D` versus `EL`. The current direct registry contains `D/CL` and `D/AL`, but not `D/EL`. It must remain labelled **proposed** until collision analysis, positive cases, exact-name guards, and a downstream rerun justify it.

3.2 All weighted spelling groups---not keyboard rules

The general weighted edit score uses nine groups. Every different ordered pair inside a group costs .45; the groups are similarity evidence, not one canonical spelling.

Group	Members	Directions	Typical readings
Consonant shape	C, K, Q	6	C↔K, K↔Q, Q↔C
Hissing sound	S, Z	2	S↔Z
Lip sound	F, V	2	F↔V
Closed-lip shape	P, B	2	P↔B
Dental sound	D, T	2	D↔T
Soft/hard sound	G, J	2	G↔J
Joined hum shape	M, N	2	M↔N
Vowel handwriting	I, E, Y	6	I↔E, E↔Y, Y↔I
Rounded vowel	O, U	2	O↔U

That is 26 weighted directions. Adding the two direct-only E/G directions gives 28 observable one-character mappings, but E/G is deliberately excluded from the global weighted metric so an incidental E/G alignment cannot disturb an unrelated multi-error ranking.

Real QWERTY evidence is separate: horizontal and vertical neighboring keys may reverse exactly one keyboard substitution. The normal keyboard generator is limited to names of length 4–16; a separate exact short path handles length 3. The older 800-case ordinary benchmark covers horizontal neighbors; the new hard suite also separates arbitrary substitution from keyboard evidence.

Why E/G is kept separate

E belongs to the I/E/Y handwriting group and G belongs to G/J. Joining the groups would incorrectly imply I, Y, E, G, and J are all interchangeable. The implementation therefore applies E/G only through bounded directional evidence and never through transitive canonicalization.

3.3 All OCR digits, ligatures, and phonetic rewrites

Registry	Every active observed ↦ catalog direction
OCR digits (9)	0↔O; 1↔I; 1↔L; 2↔Z; 3↔E; 4↔A; 5↔S; 6↔G; 8↔B. Digits 7 and 9 have no expansion.
Ligatures (14)	RN↔M; M↔RN; CL↔D; D↔CL; LI↔H; H↔LI; RI↔N; AL↔D; NN↔M; VV↔W; W↔UU; IV↔N; IA↔A; II↔U.
Phonetic (21)	PH↔F; F↔PH; CK↔K; K↔CK; X↔KS; KS↔X; CKS↔X; QU↔KW; KW↔QU; QU↔CW; GH↔G; TH↔T; TH↔S; GHT↔T; WH↔W; SH↔CH; CH↔SH; TION↔SHUN; Y↔I; I↔Y; Y↔EE.

All 14 ligature directions participate in bounded one-step exact retrieval. The mixed ligature-plus-vowel generators intentionally use only the first six directions: RN↔M, CL↔D, and LI↔H. A special N↔RI expansion exists only for exact three- and four-character family retrieval. None of these one-step outputs is recursively fed back through the same registry.

3.4 Bounded long phonetic collapse plus one ordinary error

Two long collapses need one extra, tightly bounded path because an OCR word may contain both the collapse and one ordinary type:

Rule	Concrete cases	Boundary that keeps it safe
CKS→X plus one edit	CKSEFO→XEFO; XSSACKS→ASSAX	Exact catalog lookup only; the complete family set must have at most four members.
GHT→T plus one edit	GHTAXA→TADA, TALA; OGHTA→OTAL, VOTA, YOTA	Query length 5–20; the character created by GHT→T cannot be edited again.
Rank preservation	OGHTAL→OTAL remains rank 1	The rule adds evidence and reserves hidden tail slots; it never promotes itself to rank 1 or moves an already-visible family to the tail.

This path is intentionally not generalized to every phonetic rewrite. It was added only after the hard ambiguity suite found equal protected-span evidence that the exact-only phonetic lookup could not retrieve. Algorithm 6 also protects these bounded tail rows from consensus eviction.

3.5 Skeleton, phonetic, and chain keys

Mechanism	Exact transformation	Concrete boundary
Skeleton key	PH→F; [CQK]→K; [PV]→B; [SZ]→S; remove A/E/I/O/U/Y; collapse repeats.	PAVA becomes skeleton B, while FAVA becomes FB. F is not merged with B/P/V.
Phonetic key	PH→F, CK→K, GH→G; [BPFV]→P; [DT]→T; [CGKQ]→K; [SZ]→S; J→G; remove vowels; collapse repeats.	The ordered pipeline is a coarse retrieval key; full-name edits still decide ranking.
Visual chain groups	AOUE; ILT; HNBR; MN; GQY; UVW; FTL; ECO.	Used only inside named bounded multi-operation chains, not as a global substitution cost.
Sound chain groups	BP; DT; GK; SZ; FV; CKQ.	At most the operation count allowed by the named chain generator; not an unrestricted phonetic search.
Visual-gap key	PH→F, CK→K, QU→K, GH→G, C/Q→K, repeats collapsed.	This is the separate fragment projection used by the next section; target widths stay measurable.

3.6 The 27 explicit source aliases

These are literal compact-query entries in the inherited base retriever. Each points to a source retriever target, which may be an exact base family or a broader prefix/display head. In particular, PANADOL is a source target rather than a standalone exact base family. The entries are not learned from search results and do not create a general sound-alike rule.

Alias → source target	Alias → source target	Alias → source target
BANADOL→PANADOL	BANDOL→PANADOL	PANDOL→PANADOL
PNDL→PANADOL	PANDL→PANADOL	PANADL→PANADOL
PANADOLE→PANADOL	BANADOLE→PANADOL	BANADOLCOLD→PANADOL
PANDOLCOLD→PANADOL	BANDOLCOLD→PANADOL	OGMENTIN→AUGMENTIN
OGMNTIN→AUGMENTIN	AUGMNTIN→AUGMENTIN	AUGMANTIN→AUGMENTIN
FOLTARIN→VOLTAREN	VOLTARIN→VOLTAREN	FOLTAREN→VOLTAREN
BRUFN→BRUFEN	BRFN→BRUFEN	BRUFIN→BRUFEN
BROFEN→BRUFEN	KETOFN→KETOFAN	KETOFEN→KETOFAN
NEKSium→NEXIUM	NEKSUM→NEXIUM	NEXUM→NEXIUM

3.7 Budgets that keep the OCR rules bounded

- Fewer than 4 visible name characters: no direct OCR rewrite.
- 4–5 characters: at most one registered direct confusion.
- 6–24 characters: at most two registered direct confusions.
- More than 24 characters: the direct rewrite recursion is disabled.
- The ordinary public variant-list call retains at most 512 deterministic variants. Exact-family retrieval, corrected-prefix retrieval, and the global promotion audit deliberately enumerate the full reachable set without that cap.
- Rewrites consume the original span. One letter cannot become E, then G, then another group in the same path.
- Ambiguous exact families may be retrieved and shown. Global minimum-cost uniqueness is required only when the direct-grapheme layer tries to promote one family to rank 1; that candidate must be in the first 12 and within the live .65 score gap. A literal exact catalog name is restored above confusion-only alternatives.

3.8 How a family enters the candidate pool

Channel	What the engine opens	Bound that prevents flooding
Exact family	Exact compact catalog-family key.	Literal equality is protected again after ordinary promotions.
Prefix and suffix	Prefix and reverse-suffix indexes of length 2–12.	Normal query prefix retrieval starts at 3 characters; suffix at 4.
Rare character grams	Up to four rare 4-grams; if the pool is below 160, up to five rare 3-grams.	Common buckets do not receive the same expansion.
Skeleton and phonetic	Exact keys and key prefixes of length 3–10.	A key match is retrieval evidence; the full name still supplies edit and position scores.
Delete index	No deletes below length 4; one delete at 4–7; two at 8 and above.	Retained delete key length at least 3; a query delete bucket over 650 families is not opened.
First-character expansion	E→G and G→E only, before prefix and length-index lookup.	The nine broader confusion groups stay scoring evidence; they do not expand the first-character pool.
Compatible-length scan	Query length 4–12, only when the base search is weak or uncertain. Radius 3, or 0 when the core pool already has at least 220 rows.	Plausible first character required; at most 2,200 family IDs.
Bounded nearest	Query length 3–16; radii 1, 2, 3, 4 for lengths 3, 4–6, 7–9, 10–16.	Only minimum-distance ties; at most 12 families; plausible first-character evidence required.
Short rescue	Exact length-3 keyboard neighbor; two-deletion frame for length 3–4; consonant/phonetic frame for length 3–7; equal 2-character edge; short visible head.	Short frame keeps 8, raw distance ≤ 3 , weighted distance ≤ 2.25 ; visible-head surfacing keeps 4 and is not automatic rank 1.
Family head	Shared first token of length at least 4, supported by at least two catalog families and common manufacturer evidence.	Normal head rescue uses query length 5–18 and validated head distance at most 2.
One-step evidence variants	Exactly one OCR digit, leading swap, ligature, phonetic rewrite, stuttered prefix, transposition, or keyboard neighbor.	Evidence variant limit 12; mixed ligature/transposition limit 48; normal keyboard names length 4–16.
Exact multi-operation chains	Visual+sound, two visual+delete, transpose+delete, ligature+vowel, ligature+vowel+transpose, transpose+vowel+delete, vowel+sound+delete, keyboard+vowel+delete, and short digit+vowel/sound.	Each chain has its own length, depth, transformed-distance, rank, score-gap, and uniqueness gate; the 17 promotion policies are below.
Corrected prefix	A direct rewrite becomes an exact prefix of a longer catalog family.	Query length 6–24, total cost ≤ 1.40 , prefix bucket ≤ 4 , result limit at least 2; tail visibility only, never rank 1.

Channel	What the engine opens	Bound that prevents flooding
Context-clean second pass	Remove recognized strength/unit/form noise and run the base name search again.	Cleaned name must differ and retain at least 3 characters; short weak prefixes are blocked and leading numeric brands remain.
Legacy unreadable modes	Structured <i>before, middle, and after</i> modes filter merged A5 candidates by visible anchors.	Target must be longer than visible text; this older dict mode is separate from the explicit Algorithm 6 gap grammar in the next section.

3.9 Edit costs, prefilter, and full family score

Observed operation	Cost	Important scope
Insertion; deletion; ordinary substitution	1.00	Default raw spelling error.
One of the 26 weighted-group directions	.45	Global weighted comparison.
Other vowel-to-vowel substitution	.70	Used only when the pair did not already receive .45.
Adjacent transposition	.55	Raw Damerau still counts one edit.
Listed OCR digit→letter	.45	OCR-visual comparison only.
Direct CL/D, AL/D, I/E/Y, or E/G operation	Registry cost	Bounded direct evidence; E/G is not a global discount.
Repeated arbitrary character	deletion cost	Only a duplicated 2- or 3-letter prefix has a special exact stutter rule.

The broad prefilter is deterministic and bounded. With prefix P , suffix S , character-gram Jaccards G_2, G_3, G_4 , skeleton/phonetic similarities K_s, K_p , subsequence Q , weighted edit E_w , position X , and length coverage C :

$$S_{\text{pre}} = 1.35P + .65S + .55G_2 + 1.20G_3 + 1.00G_4 + .70K_s + .65K_p + .35Q + .90E_w + .45X + .25C.$$

The main shortlist keeps 45 family rows; the edge path reserves 15. A full candidate is then scored as

$$S_{\text{family}} = 1.25I + .58E + .52E_w + .16P + .10S + .10G_2 + .18G_3 + .16K_s + .14K_p + .10Q + .24X + .16C,$$

plus bounded first-character, close-length, prefix, and weighted-position bonuses and weak-short penalties. Normal admission is at .62. The narrow positional rescue may enter from .58 only with the same first character, position agreement at least .40, and length coverage at least .70.

The original base search, cleaned-context pass, and rescue search are merged with different rank-aware contributions:

$$S_{\text{base}} = .72 \min(s, 1) + \frac{.32}{r+2}, \quad S_{\text{context}} = .84 \min(s, 1) + \frac{.40}{r+2} + .05, \quad S_{\text{rescue}} = s + \frac{.18}{r+1}.$$

Non-confident base evidence is multiplied by .90; base/context agreement adds .06 and base/rescue agreement adds .08. Those scores form a candidate order, then only the named gates below may change it.

3.10 When a candidate may move up: every promotion family

What "promotion" means

A promotion is a **bounded reorder**, not a confident medicine selection. The family must already have been retrieved, and every condition of one named gate must pass; otherwise the order stays unchanged. Here, "rank" means the family's position immediately before that gate runs—not its Algorithm 2 external rank or rescue rank. Literal exact-name restoration runs after the ordinary promotion chain, and every final result still requires confirmation.

Promotion family	What may move	Main blocker
Four core corrections	Dual closer; pure deletion; closer single-token over multi-token; validated variant head.	Exact top or any failed raw, score, token, group, or head-distance conjunct.
Unique and contained nearest	Unique first-three candidate with raw ≤ 4 , gain ≥ 2 , gap $\leq .75$; or rank-2 contained candidate raw ≤ 1 , gain 1, gap .25.	Equal nearest families remain ambiguous.
Preserved-top nearest release	Rank 2 passes a preserved top only with the required raw, weighted, edge, and score dominance.	A tie or missing strict improvement keeps the incumbent.
Exact ligature	Unique exact ligature evidence among the first five.	New variable-length direct candidates are limited to original ranks 2–3; raw-worse $AL \rightarrow D$ is blocked.
Weighted edge	Unique tied/near rank-2 or rank-3 family with weighted gain (.50 or .70), edge gain, and bounded gap (.25 or .40).	Full-name and historical regression guards block broad visual overpromotion.
Exact-key Pareto	Exact skeleton/phonetic candidate no worse on weighted, position, edge, and dual evidence; rank at most 3, gap at most .15.	Nonunique key or any worse signal.
Transposition	Unique exact one-swap candidate at rank 2–3, or a dual rank-2 extension, with raw distance at most 1 and bounded gap.	Short/nonunique swap or worse weighted evidence.
Keyboard	Exact one-key neighbor at rank 2; weighted extension requires gain at least .25 and gap at most .15.	Cannot erase an equally supported collision.
Narrow tie breakers	Exact visual edge, phonetic-position, shifted-edge, visual-distance, and skeleton-position evidence.	Each requires its own unique edge/position/key gain and a score gap between .05 and .40.
Score-chain release	A chain candidate already stronger on bounded score/raw/weighted evidence may be released from a preserved top.	Retrieval reason alone is never enough.
Stutter prefix	Removing one duplicated initial 2- or 3-character block produces exactly one catalog family.	Multiple targets or arbitrary internal repeats do not receive this rule.
Exact phonetic rewrite	One exact one-step phonetic family inside the first 5.	Competing sound-alikes remain visible rather than becoming false uniqueness.
Strict full name	Rescue-only family, query length at least 5, raw at most 3, weighted gain at least .35, edge gain .05, rescue rank at most 5, gap .35.	Exact/prefix/contained incumbents and failed edge evidence.
Family-head and character Pareto	Validated family head, short visible head, ordered-character head, bounded head, or Pareto-character evidence.	Shared manufacturer, LCS, edit, edge, score, uniqueness, and full-name guards.
Two-sided anchor	Unique family supported at both name edges, gap at most .40, with weighted/LCS/edge checks.	Equal support does not reorder.
Direct grapheme	Globally unique minimum-cost exact target within rank 12, live gap at most .65, weighted no worse, normally raw disadvantage at most 1.	Literal exact query, ambiguity, transitive path, raw-worse $AL \rightarrow D$, or any failed conjunct.
Exact-name restoration	Literal exact compact catalog family returns above confusion-only alternatives.	Legacy unreadable mode is a separate branch.
Corrected-prefix/post-repair	Longer corrected-prefix families are tail surfaced; three narrow repairs protect a strict Pareto chain, an external-rank tie, or one constrained top-5 chain family.	No general rank-1 jump.

Promotion family	What may move	Main blocker
Algorithm 6 consensus	Algorithm 5 stays the base order; the learned rank-1 gate is disabled. External families need 3 sources, Levenshtein .55, and Jaro–Winkler .85.	One replacement if A5 is full; only vacant slots are filled if it is under-full.
Final decision	Every returned family remains low-confidence decision support.	<code>needs_clarification=true</code> and <code>confirmation_required=true</code> .

Worked examples from frozen tests. These rows show gates that actually fired in the preserved final run; they illustrate the rules above rather than adding new rules.

Promotion family	Observed input $\mapsto$ expected family	Frozen result	Why the move was allowed
Unique nearest	VVEIORN $\mapsto$ WE IORN	Expected family finished rank 1	One unique near family; raw distance at most 4, gain at least 2, and gap at most .75.
Exact ligature	RIEOMUNE $\mapsto$ NEO MUNE	Expected family finished rank 1	One exact ligature target, also the unique nearest family, with dual retrieval and no worse weighted distance.
Weighted edge	CLMAE $\mapsto$ DMAE	Expected family finished rank 1	Raw-distance tie, but DMAE was the unique weighted winner and had sufficient edge evidence.
Strict full name	MELTAOIVIN $\mapsto$ MELATONIN	Expected family finished rank 1	Rescue-only target with raw distance 3 or less plus weighted and edge gains; no exact/prefix incumbent blocked it.
Validated family head	UROSOLVINYN $\mapsto$ URO-SOLVINE N	Expected family finished rank 1	One validated family head was within two edits, closer than the incumbent, and began with the same character.
Direct grapheme	CACLEX $\mapsto$ CADEX	Expected family finished rank 1	Globally unique minimum-cost exact target, already within rank 12, with gap at most .65 and no worse weighted distance.

The blocker is part of the rule

MYLANO can reach both MELANO and MILANO at equal minimum cost, so both must remain visible and neither receives a deterministic grapheme promotion. Conversely, literal CEFIX must remain rank 1 even when a confusion-only alternative exists. A positive example without its ambiguity and exact-name guards would be an incomplete description of the rule.

3.11 The 17 exact-chain promotion policies

Each row starts only after its operation chain reaches an exact catalog key. Let r be current rank, $g = S_{\text{top}} - S_{\text{candidate}}$, and d be the candidate raw-distance disadvantage. “Exact key” means a skeleton/phonetic advantage; “dual” means base and rescue support.

ID	Chain	Rank/gap/raw bound	Additional evidence
1	Ligature+vowel+transpose, core	$r \leq 5, g \leq .40, d \leq 1$	Unique; normal head protection.
2	Ligature+vowel+transpose, extension	$r \leq 5, g \leq .75, d \leq 1$	Preserve correction; relaxed head protection.
3	Ligature+vowel, core	$r \leq 5, g \leq .40, d \leq 0$	Unique; candidate cannot be raw worse.
4	Ligature+vowel, distance extension	$r = 2, g \leq .05, d \leq 1$	Preserve an existing correction.
5	Ligature+vowel, edge extension	$r = 2, g \leq .15, d \leq 1$	Edge no worse; preserve correction.
6	Ligature+vowel, multi-chain extension	$r \leq 5, g \leq .40, d \leq 1$	At least two exact-chain reasons.
7	Two visual steps + one sound step	$r \leq 5, g \leq .10, d \leq 1$	Unique and dual support.
8	Visual+sound exact-key extension	$r = 2, g \leq .40, d \leq 0$	Exact key and weighted no worse.

ID	Chain	Rank/gap/raw bound	Additional evidence
9	Visual+sound dual extension	$r \leq 4, g \leq .15, d \leq 0$	Dual, weighted no worse, relaxed head protection.
10	Keyboard+vowel+delete, core	$r \leq 5, g \leq 0, d \leq 1$	Exact-key advantage.
11	Keyboard+vowel+delete, extension	$r \leq 3, g \leq .25, d \leq 1$	Exact key and weighted no worse.
12	Two visual steps + target delete	$r \leq 3, g \leq .02, d \leq 1$	Position no worse.
13	Visual+delete multi-chain	$r \leq 3, g \leq .15, d \leq 0$	Weighted no worse; at least three chain reasons.
14	Transposition+target delete	$r \leq 3, g \leq .25, d \leq 0$	Exact key and weighted no worse.
15	Transposition+vowel+delete	$r \leq 5, g \leq 0, d \leq 0$	Preserve correction; relaxed head protection.
16	Vowel+sound+delete, short	$r = 2, g \leq .25, d \leq 2$	Query length 3–4; weighted no worse; two reasons.
17	Vowel+sound+delete, multi-extension	$r = 2, g \leq .10, d \leq 2$	Query length 3–4; exact key; two reasons.

The list describes promotion policies, not every retrieval route. Short OCR+vowel/sound and the short two-deletion frame also retrieve exact families, but they flow through the short/evidence and baseline-preservation gates.

4 Visual-gap rules

The visual-gap mode is for a user who can read fragments but not the text between them. It is a distinct query grammar, not a general typo symbol.

Mode	Example	Meaning
Leading gap	...DOL	At least one hidden target character before DOL.
Trailing gap	PANA...	At least one hidden target character after PANA.
Internal gap	PANA...DOL	PANA and DOL occur in order with at least one real hidden target character between them.
Both edges	...INADC...	Hidden text exists at both edges; the exact family CLIN-DACIN ADCO is recoverable.
Multiple gaps	DI...CY...ONE	Up to four retained fragments, in order.
Whitespace short-hand	PANA DOL	Two to four alphabetic fragments; joining may be zero-width.

Accepted explicit markers

Two or more dots, the ellipsis character, one or more asterisks, one or more question marks, or two or more underscores. Punctuation-separated words that do not match this grammar stay ordinary spelling queries. A **single** * or ? is therefore an active gap marker; this exact boundary is covered by the generated marker-glyph tests.

4.1 Four matching stages

1. **Literal ordered fragments:** exact visible text in the required order.
2. **Fixed grapheme projection:** PH/F, CK/K, QU/K, GH/G, C/Q/K, repeats.
3. **Bounded OCR confusion:** the direct registry above, active only for 5–24 visible characters; one confusion below 8 visible characters and at most two from 8 upward, with cost budgets 1.00 and 1.40.
4. **One ordinary visible edit:** insertion, deletion, or substitution only when total visible text is at least five characters and every fragment has at least two characters.

Every explicit marker must correspond to at least one target character. The matcher backtracks across repeated fragment occurrences, preserves the exact matched base family, and reports hidden characters and coverage on the target side. All returned rows require confirmation.

Test	Expected behavior	Rule being protected
AT...OL	ATENOLOL is recoverable using the final OL, not the first one	repeated-occurrence backtracking
PANA...DOL vs PANADOL	exact PANADOL is rejected	internal marker must hide a character
...PANADOL vs PANADOL	exact PANADOL is rejected	leading marker must hide a character
BRU... + 600 mg	exact BRUFEN base is selected, not a broad sibling group	visual exact-family identity
VIT__3	visual mode stays visual; bare 3 is not silently parsed as product context	cross-mode isolation

4.2 What the new visual benchmark actually tests

Cohort	Rows	What is varied	Example
Structural positives	240	Leading, trailing, internal, both-edge, multi-fragment, and whitespace shorthand; 40 each	...AGLOBE ↦ DEXAGLOBE; LUM OIL ↦ LUMIERE HAIR OIL
OCR/glyph composition	54	One OCR confusion plus a gap; every accepted marker glyph	WOMYGR... ↦ WOM- EGRANTY
Numeric visual names	20	Digits belong to the medicine name and must not become product strength	VITA...D3K2 ↦ VITAMIN D3 K2
Exact ambiguity	30	Every exact base family satisfying the same mask is labelled relevant	PRE...RAX keeps PREGO-RAX and PRETERAX
Safety boundaries	80	Zero-width explicit gaps, short-confusion floor, punctuation collisions, and numeric shorthand isolation	CEVI...ANO must not match exact CEVIANO

The 424 rows were generated before execution from exact catalog base-family surfaces and split by family-stable hashes (338 development, 86 holdout). The oracle stores both the selected exact source family and the complete set of exact families satisfying the mask. Evaluation reads `matched_family_key`; the broad display group never earns credit.

5 Product-detail rules

Optional product details never replace the medicine-name evidence. They filter or reorder products only inside a bounded set of plausible exact families.

5.1 Numeric evidence

Rule	What is recognized	Safety behavior
Qualified strength	mg, mcg/ug, g, kg, IU/U, MIU, percent, and safe unit conversions	Explicit zero or malformed magnitude fails closed.
Concentration	numerator/denominator including mL, L, g, microlitre, and time denominators	Exact printed tuple wins; a scaled-equivalent ratio is compatible but not an equal exact tie.
Combination strength	repeated components, shared denominators, mixture ratios, colon ratios	All supplied components must agree.
Presentation quantity	container weight/volume such as 50 g cream	Quantity alone is weak; quantity plus matching structural form may select.
Bare number	600 with a plausible family; or number plus form	May express strength or pack count, but only through bounded hypotheses.
Package count	tablets, capsules, doses, actuations, patches, suppositories, pens, syringes, cartridges, multipacks	Every supplied count must be satisfied; one matching count cannot hide an extra conflicting count.

5.2 Form, route, release, and container

- Form aliases distinguish tablet, capsule, liquid, cream, gel, ointment, drops, spray, suppository, solution, vial, ampoule, syringe, pen, cartridge, and other catalog forms.
- Route keeps oral/topical/ophthalmic/etc. evidence and preserves injection subtypes such as IV, IM, and SC when explicitly supplied.
- Release terms distinguish immediate products from SR, XR, ER, CR, MR, retard, extended-release, and dotted abbreviations.
- If the query supplies a category and the product lacks or contradicts that metadata, the product is not labelled compatible.

5.3 Family admission and selection

Situation	Active rule
Exact medicine name	Hard boundary: product context cannot jump to another base family or a sibling folded into a broad display group.
Ordinary nonexact name	Rank 1 is admitted; ranks 2–3 require every available diagnostic to pass raw distance ≤ 2 and similarity $\geq .60$; rank 4+ is excluded.
Visual-gap name	First three structurally matched exact base families are eligible; broad variant siblings are not opened automatically.
Strong exact numeric evidence	Compatible products can reorder admitted families; incompatible name-only rows do not leak into a successful product selection.
Weak form/presentation evidence	May filter, but cannot freely pull a distant medicine above a stronger name match.
No compatible product	Return the admitted medicine name(s) with an explicit no-compatible decision; never invent a product.
Product tie	Keep every genuinely tied stable product ID, subject to the public result limit.

Three examples that define the intended behavior

```
javaki + 5 mg → rank-2 name plus exact strength → JAKAVI 5 MG
brufen + 600 → exact family lock plus strength → three legitimate BRUFEN 600 MG ties
brufen + 600 tab → strength plus form/package evidence → one compatible BRUFEN tablet product
```

5.4 Numeric aliases and very short prefixes

Exact commercial aliases beginning with numbers are indexed by exact base family. The longest exact alias wins, an unknown suffix blocks the rescue, and visual-gap mode disables numeric-alias and short-prefix shortcuts. The special numeric-only alias `1 2 3` intentionally returns both exact families and requires confirmation. A one- or two-letter prefix such as `x + 500 tab` may use context only when the family set is small and the numeric/form evidence is compatible; a broad or conflicting set abstains.

6 Safety rules around every result

Guard	What it prevents
Literal exact-name protection	A confused spelling cannot outrank the exact catalog spelling entered by the user.
Global uniqueness before promotion	An OCR rewrite shared by multiple families cannot become a confident single correction.
Original-span non-transitivity	Overlapping visual groups cannot chain one character through several identities.
Exact base-family identity	Product details and visual matches cannot open unrelated siblings from a broad display group.
All supplied details must match	One correct strength or count cannot hide a conflicting route, release, container, or second count.
Stable product IDs	Duplicate commercial-name strings remain distinct products with their own price/-company/record.
Visual-mode isolation	Gap punctuation cannot be cached or normalized into an ordinary/numeric-alias query.
Always confirm	Every result is decision support, never autonomous clinical selection.

7 The new test sets

The selected name-rule cases below are organized as five separate claims. The complete 336-row registry remains the source-to-test crosswalk for rules that could not enter these catalog-derived files. Accuracy misses, hard safety failures, broad collision diagnostics, and source-boundary failures are never pooled into one reassuring percentage.

3,284

hard name-accuracy rows

272

hard ambiguity/exact guards

100+11

diagnostic + source rows

7.1 Hard name-reading standard: five frozen files

File	Rows	Verdict type	What it isolates
name_reading_atomic_accuracy.csv	1,724	Accuracy	One controlled mapping, one missing ordinary-error channel, or one declared alias.
name_reading_composed_accuracy.csv	1,560	Accuracy	Exactly two non-overlapping operations across 13 composition signatures.
name_reading_collision_safety.csv	272	Hard gate	100 atomic query collisions, 100 same-fingerprint composed collisions, and 72 literal exact-name guards.
name_reading_cross_channel_diagnostics.csv	100	Diagnostic	A broader query collision reached through unlike two-operation paths; relevant breadth is measured but not mislabelled equal evidence.
name_reading_source_boundaries.csv	11	Hard gate	Length/depth, non-transitivity, E/G-only first-character scope, deterministic order, and the default 512-variant cap.

Four difficulty levels, four different claims

L1 atomic and **L2 composed** report Hit@1, Hit@5, Hit@20, and MRR; every miss stays in the denominator. **L3 strict ambiguity** must show every family generated by the same declared evidence fingerprint, while a literal exact query must remain rank 1. Cross-rule L3 collisions are breadth diagnostics because their paths and costs differ. **L4 source boundaries** are direct pass/fail contracts. Wrong runtime identity or missing confirmation invalidates any row regardless of rank.

7.2 What “hard” means in this suite

Coverage layer	Exact construction
Source inventory	71 deduplicated observed→catalog mappings from the weighted groups, direct graphemes, OCR digits, ligatures, and phonetic registry; 27 declared source aliases; and four previously underrepresented ordinary channels: arbitrary insertion, general substitution, whitespace, and punctuation.
Selected hard rows	The frozen files contain 70 mapping IDs and 16 aliases. TION→SHUN had no eligible exact-family witness; the 11 PANADOL-target aliases point to a prefix/display target rather than a standalone exact base family. Those omissions remain visible in the registry instead of being misreported as tested rows.
Atomic universe	482,315 independently enumerated candidates and 482,048 distinct queries before quota selection; no search rank is read during generation.
Composed universe	540,465 candidates and 539,960 distinct queries. Each selected row applies two non-overlapping operations to one exact catalog family.
Cross-rule composition	Registered rule plus deletion, duplicate, substitution, insertion, spacing, or transposition; same registered rule twice; two different registered rules; deletion+transposition; duplicate+deletion; insertion+substitution; spacing+substitution; two substitutions.
Leakage control	Every family linked by any generated query collision is joined into one of 17,118 connected components; SHA-256 assigns the whole component to development or holdout.
Failure preservation	Every run snapshots its manifest and exact CSVs, then stores per-case rows, failures, summary, evaluator hash, and runtime source provenance. Bulky raw API responses go to local archival output.

7.3 Worked test cases: what one row contains and how it is judged

One test case is a frozen contract

It contains five things: **the exact input**, **the expected family or product**, **how the input was constructed**, **the judging policy**, and **the allowed rank or safety outcome**. Expected answers are fixed from the catalog and rule before the API runs—never copied from the API winner. Finding one correct family can still fail when the row requires every relevant family, forbids a family, or requires abstention.

One complete example, field by field.

Field	Frozen value	What it means
Input	GHTAXA	The exact text sent to the search API.
Construction	GHT→T plus one ordinary substitution	The same two-operation fingerprint can lead back to more than one exact catalog family.
Expected result	TADA; TALA	Both labels were known before the search ran; the semicolon means two required families, not two guesses made afterward.
Judging policy	All relevant families visible within top 20	Returning only TADA or only TALA fails even if that one result is rank 1.
Safety outcome	Ambiguous; confirmation required	A confident single winner fails because the evidence is genuinely nonunique.

How the main name-test types are scored.

Test type	Input $\mapsto$ expected result	Exact pass or measurement rule
L1 single rule	LAMICTD $\mapsto$ LAMICTAL	Record the first relevant rank. Rank 1, 5, and 20 feed Hit@1/5/20 and MRR; a miss stays in the denominator.
L2 composed rules	F3RROSWA8 $\mapsto$ FERROSWAB	Apply 3 $\mapsto$ E and 8 $\mapsto$ B on different original spans, then measure the frozen family exactly like an L1 accuracy row.
L3 ambiguity safety	BETAXINE $\mapsto$ BETADINE; BETA-SINE	Both independently relevant exact families must appear within top 20 and the response must require confirmation.
L3 exact-name guard	REPO $\mapsto$ REPO at rank 1	Literal REPO must stay first; confusion-reachable REBO is not allowed to replace it.
L4 source boundary	IARDX $\mapsto$ GARDX forbidden	The generator/helper must not apply I $\mapsto$ E and then E $\mapsto$ G to the same original character. This is a hard source contract, not an accuracy rank.

The same contract idea in the other test sets.

Stream	Input $\mapsto$ expected result	What passes
Visual gap	...INADC... $\mapsto$ CLINDACIN ADCO	The exact matched-family identity must appear within top 20, both explicit edge gaps must hide at least one character, and the response must require confirmation.
Product selection	brufen + 600 tab $\mapsto$ BRUFEN 600 MG 30 TABS.	The stable product ID for the compatible tablet is selected; returning only the BRUFEN family name is not enough.
Product abstention	xanax + 500 mg tab $\mapsto$ no compatible product	No product ID may be selected and the decision must explicitly report <code>product_context_no_compatible_product</code> .

7.4 Generation protocol without result leakage

1. Freeze exact Algorithm 5, Algorithm 6, catalog, generator, and evaluator bytes before looking at results.
2. Enumerate transformations over 15,795 eligible alphabetic exact catalog families. Expected labels come from the source family, never the API winner.
3. Build the full query-to-family graph before sampling; assign connected components, not rows, to development or holdout.
4. Select deterministic quotas across rule, composition, position, length, and split. Accuracy queries require one exact generated label; strict ambiguity stores the complete same-evidence relevance set.
5. Evaluate every row through the public Algorithm 6 API. Accuracy misses lower the metric; safety/source failures block acceptance; broad cross-rule collisions report recall without pretending paths are equivalent.
6. If the oracle, evaluator, or algorithm fails, keep the original run, classify the cause, apply only the justified repair, and rerun the entire affected stratum plus neighboring regression sets.

What this still cannot prove

The component split prevents generated-neighborhood leakage, but this remains a retrospective catalog-derived engineering benchmark. It is not a sealed collection of unseen prescription images or handwriting. A truly blind natural-OCR cohort needs independently collected images, family/image-level splits, dual adjudication, and frozen labels before the first test run.

7.5 Other evidence streams, kept separate

Dataset	Rows	Measures	How it was made
Ordinary typo accuracy	800	Hit@1/5/20 and MRR	Independent catalog mutation channel; balanced deletion, transposition, duplicate, keyboard neighbor; family-disjoint split.
Ordinary ambiguity safety	150	all relevant families visible; exact guards	Inverted mutation collisions plus exact catalog-name collisions; excluded from accuracy denominator.
Reviewed fair OCR	412	Hit@1/5/20 and MRR	Human-reviewed locked OCR strings; current source rerun with exact-family relevance.
Directional OCR generated	160	per-direction retrieval and depth	Invert each registry direction independently; retain globally unique minimum-cost labels.
OCR challenge/guards	78	hard regression and safety	Named hard cases, literal-name collisions, ambiguity, non-transitivity, and length boundary.
Visual-gap benchmark	424	source-family Hit@k, exact-label recall, contracts	Catalog masks generated before execution; full exact relevance set, 16 marker/mode/safety strata, family-stable split.
Visual safety	100	parser, edge, collision, and cross-mode contracts	Zero-gap negatives, ambiguous masks, short floors, marker collisions, numeric visual names, and performance bounds.
Strict product selection	440	exact product-ID set or correct abstention	Independent catalog oracle: unique strength, ties, same-strength form, impossible strength.

7.6 The row recipes, in plain language

Set	Exact composition	A representative row means
Ordinary typo	200 deletions, 200 adjacent transpositions, 200 duplicated characters, 200 horizontal-keyboard neighbors; plus 100 ambiguity and 50 exact-name guards	<code>HME_{LIS}</code> was created from <code>HAME_{LIS}</code> by one locked deletion channel; every inverse-collision family was known before search ran.
OCR registry	10 rows for each of 12 directional mappings (120), plus 40 two-confusion rows; 78 named challenge/source guards	Each arrow such as <code>G→E</code> is tested independently, then combined within the real one/two-operation budget.
Reviewed OCR	412 accepted rows: 324 development, 88 holdout, 42 with multiple acceptable exact families	A miss stays in the denominator; the scorer accepts any predeclared exact relevant family.
Visual gap	424 rows across the five cohorts shown in Section 4.2; 338 development and 86 holdout	Exact source and all exact mask matches are fixed before the API response is seen.
Strict product	240 unique strengths, 40 legitimate strength ties, 80 same-strength form disambiguations, 80 impossible-strength abstentions	The expected answer is an exact stable product-ID set, or explicitly no selected product.

What these datasets do not prove

Catalog-derived development/holdout rows are retrospective engineering benchmarks, not a blind external clinical study. A regression-set pass proves the named contract only. The fair 412 set measures reviewed OCR retrieval; it does not measure product selection. No percentage in this document is a prescribing or patient-safety claim.

8 Current-source results

8.1 New hard name-reading benchmark

1724/1724
atomic Hit@20

1556/1560
composed Hit@20

272/272
strict safety

Stratum	Cases	Hit@1	Hit@5	Hit@20	MRR@20 / claim
L1 atomic accuracy	1724	1717	1721	1724	0.997124; one declared rule or missing ordinary channel
L2 composed accuracy	1560	1504	1549	1556	0.976647; exactly two non-overlapping operations

The release gates pass 272/272 strict ambiguity and exact-name rows plus 11/11 source-boundary contracts. The separate cross-channel diagnostic finds 190/201 declared relevant labels within the top 20 and has complete visibility on 92/100 queries. Those diagnostic numbers are deliberately not converted into a safety pass.

8.2 Accuracy benchmarks

799/800
ordinary typo Hit@1

234/412
reviewed OCR Hit@1

313/314
visual source-family Hit@1

Benchmark	Cases	Hit@1	Hit@5	Hit@20	MRR / interpretation
Ordinary typo	800	799	800	800	0.999375
Reviewed fair OCR	412	234	296	340	0.633907
Visual source recovery	314	313	314	314	0.998408; preselected exact source family

The visual benchmark also passed 424/424 full contracts, found 400/400 exact oracle identities, and passed 100/100 safety rows. It returned 1410 additional bounded-fuzzy identities, so the literal exact-oracle precision indicator is only 22.1%. That is a candidate-breadth diagnostic—not a clinical precision claim and not automatic proof that every extra family is wrong.

8.3 Safety and strict selection

440/440
strict product cases

150/150
ordinary ambiguity guards

100/100
visual safety contracts

The strict product result requires the exact stable product-ID set—not just the right family name. An impossible strength passes only when the API selects no product and reports the no-compatible-product decision.

The separate deterministic 200-case name-plus-context benchmark moved from 197/200 baseline Hit@1 to 200/200 with zero regressions. Its three recoveries were `LEAL + 100G` $\mapsto$ `LEIL`, `DIAOL + 2%;120ML` $\mapsto$ `DIKOL`, and `ARGATEX + 50G` $\mapsto$ `ARGOTEX`. The focused `javaki + 5mg` case is a separate named regression, not one of those three generated recoveries.

8.4 Developer regression gates---reported separately

Gate	Latest result	What a pass means
Unified OCR/legacy visual contracts	642/642	236 API OCR rows, 2 source guards, and 404 legacy visual rows still satisfy their named contracts.
Product parser/reranker unit suite	132/132	Strength, concentration, package, form, route, re-release, alias, conflict, and identity regressions remain fixed.
Focused OCR source suite	23 positives, 2 locked regressions, 23 clean/fair safety, 21 exact guards, 7 ambiguity guards, 4 long-collapse cases, and 5 collapse rank-one guards	Direct registry examples, exact-name protection, ambiguity, non-transitivity, input bounds, and the bounded CKS/GHT repair remain intact.
Focused visual source suite	64 generated + 20 hard + 9 edge + 3 internal + 4 collision + 8 short guards	Visual grammar and boundary regressions remain intact; the heavy case stayed inside the 512-pattern cap.
API and browser boundaries	37 searches + runtime + health; JavaScript suite pass	Public request/response and client behavior still match the named scenarios.

These are regression contracts, not additional accuracy denominators. Passing the same named rows again shows that a repair did not break their expected behavior; it does not estimate performance on unseen prescriptions.

8.5 Fair comparisons only

Method on the same 412 rows	Hit@1	Hit@5	Hit@20	MRR@20
Current Algorithm 6	234	296	340	0.633907
Current Algorithm 5	234	296	338	0.633664
Damerau–Levenshtein	179	251	299	0.51581
Jaro–Winkler	149	256	312	0.478935

All four use the same 412 rows, exact-family scorer, top-20 limit, and 17,476 family universe. The lexical rows are retrieval references, not equivalent product systems: they do not implement Algorithm 6 visual semantics, safety, consensus, product selection, or confirmation. Algorithm 6 and Algorithm 5 tie at Hit@1/5; Algorithm 6 adds two Hit@20 recoveries with no losses: LACTULOSE $\mapsto$ LACTO and TEFA $\mapsto$ TELFAST.

The older 464-row and current 412-row percentages are not an algorithm improvement: the 52 reviewed exclusions were all old top-20 misses, so the numerator stayed the same while the denominator changed. They are separate protocol versions.

9 What failed, what changed, and what was rerun

9.1 Hard name benchmark: protocol failure, then runtime failure

The first preserved run passed only 264/272 safety rows. The generator had grouped families reached by unlike two-operation paths as if they had equal evidence. That was an oracle defect: the paths and costs differed. The hard set was rebuilt around the complete operation fingerprint, while the broader collisions moved to a diagnostic set instead of being deleted.

The second preserved run passed 269/271 strict rows and exposed two genuine failures, `GHTAXA` and `OGHTA`. Independent inverse-operation enumeration also showed that the latter oracle was missing `YOTA`. The runtime received the bounded CKS/GHT-plus-one-edit evidence path described above, and the oracle was completed independently. A first tail-surfacing attempt broke `OGHTAL` $\mapsto$ `OTAL` rank 1; that implementation was rejected and repaired before acceptance.

The final current-source run is 20260822T225802Z: safety 272/272, boundaries 11/11, and the accuracy numbers shown above. The original failures, raw-response hashes, source hashes, and both repairs remain in the append-only failure ledger.

9.2 Strict product benchmark: 435/440 first run

Five first-run failures were preserved. Investigation showed that the simple catalog oracle had trusted an incomplete `st` field while the printed commercial name contained a combination or concentration denominator. That mislabelled LEVOXIN, CAZET, LIPITRIN, CHOLEROSE, and EXAMIDE cases.

Classification	Root cause	Repair and reevaluation
Label/oracle defect	Composite sibling meta-data, printed multi-strength combinations, or a printed denominator absent from the simple <code>st</code> field	Exclude the whole unsafe family from this simple-evidence stratum; keep the original failed artifacts; regenerate by the same seed; rerun all 440 rows.
Algorithm change	None justified	Final strict result: 440/440.

9.3 Visual benchmark: 422/424 first run

The two initial failures used a mistaken oracle assumption that one `*` or one `?` was ordinary punctuation. Source inspection showed that both are explicit markers in the current parser. The category labels were corrected, the original run was preserved, and all 424 cases were rerun. Runtime source hashes stayed unchanged; this was an oracle repair, not an algorithm change.

9.4 Failure policy for every future rule

1. Freeze source, catalog, generator, evaluator, and dataset hashes.
2. Preserve the raw failing response and exact row.
3. Classify the cause: algorithm, label/oracle, evaluator, API/display, UI, or environment.
4. Make the narrowest justified change. Never delete a case just because the algorithm missed it.
5. Rerun the failed stratum, its safety neighbors, and the downstream non-regression sets.
6. Put the limitation beside the final number.

10 Open team decisions

1. Should D/EL become a real directional OCR rule? It still needs catalog collision analysis and positive/negative evidence.
2. Which additional handwriting pairs have real annotated examples rather than intuition alone?
3. Who will supply a genuinely blind external image/OCR set? Internally generated rows cannot answer external generalization.
4. Which product metadata gaps should be corrected in the catalog rather than accommodated by broader parser logic?

Companion evidence

The PDF is intentionally readable. The accompanying folder contains the complete 336-row source rule registry, all CSV test rows, manifests, row-level results, compact summaries, and failure history. Content-addressed raw API logs remain reproducible local run artifacts rather than Git payload. That material is the audit trail; this handbook is the team explanation.

A Rule-to-test checklist for contributors

Every proposed rule must arrive with the following minimum evidence before it can move from proposed to active:

Required evidence	Question answered
Positive examples	Does the intended difficult input recover the correct exact family/product?
Negative example	Does a nearby but unsupported input remain unsupported?
Exact-name guard	Does a literal catalog name remain rank 1?
Ambiguity/collision guard	Are all equally supported families kept visible and confirmation-required?
Boundary cases	Do the exact length, cost, rank, and operation limits behave on both sides?
Cross-rule case	Does the new rule remain safe with visual gaps, product details, or numeric aliases?
Downstream rerun	Did any previously correct case lose Hit@1/5/20 or strict product identity?
Failure note	If anything failed, is the original result, cause, repair, and rerun preserved?

B Current authority identifiers

Component	SHA-256 prefix used in this handbook
Algorithm 5	c5faefb2...
Algorithm 6	cedf1fce...
Product-context reranker	03780539...
API adapter	752399ee...
Catalog	d234edaa...

C Dataset and run identifiers

Dataset	Rows	SHA-256 prefix	Preserved current run
Ordinary typo accuracy	800	5c00ce528f06	20260822T230616Z
Reviewed fair OCR	412	3ad1a423cadc	20260822T230528Z
Visual exact-identity	424	01d2aa319edb	20260822T230942Z
Strict product selection	440	ce3d09e3446c	20260822T230841Z
Unified generated/challenge regression	642	multiple locked CSVs	20260822T230951Z
Hard name-reading suite	3667	48c1df5ec62f	20260822T225802Z

The machine-readable evidence bundle contains the full hashes, paths, source identity, preserved metrics, and claim boundaries. The generated LaTeX macro file is rebuilt from that bundle's validated inputs; numbers are not typed manually into this handbook.